Technical Articles Theme 07

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What is innovation?

Innovation, creativity and entrepreneurship are closely related. Innovation is defined as "the process of translating an idea or invention into a good or service that creates value for which customer will pay." In large organizations, it is a complex process to bring innovation because of certain limitations. On the other hand, small firms are more prone to innovation and every small startup is thought to be innovative. The reason is; less complex structure of small firms, quick decision making, easy resource pulling, and easier product testing.

Innovation is also defined as: "Making a product or service better, faster, cheaper, greater satisfaction and/or greater value." Another comprehensive definition of innovation is "the company's capability to implement a range of coordinated actions and strategies for the purpose of developing and delivering new products and/or new services to the new or existing market in a manner that outperforms competitors." Two critical components of innovation are: invention and commercialization. Commercialization adds social and economic value to the invention through the creation of a business, or/and licensing/intellectual property protection. You need to commercialize the new idea to solve the society's problem. Without commercialization, no matter how unique the idea is, it won't be converted into a product and offered in the market. The most common types of innovation are: incremental innovation, disruptive innovation, architectural innovation and radical innovation.

Most of the firms go for **incremental innovation** with existing market and existing technology. Incremental innovation is referred to the series of improvements a company make in its existing products and services with time. For example, Apple or Samsung mobile companies launch a new model of their smartphones every year with a slight improvement in certain features. Incremental innovation is not risky. **Disruptive Innovation** is related with implementing new processes, technologies (disruptive business models) in existing markets. In the beginning, new models look inferior as compared to the current solutions, but with the passage of time, they surpass the old technology due to their efficiency and efficacy. Amazon was the first online book store which disrupted the old business model of conventional book stores. There was an existing market of books and they changed the way books are sold. **Architectural innovation** is what the giant companies like Google and Amazon do. They take their domain related technology, expertise and skills and then implement it to a different market. This is the way they increase their customer base by expanding the market. **Radical innovation** is the type of innovation which people actually think as innovation stereotypically. It is the rarest type of innovation. It involves innovating new technologies, business models or services which create new markets. The examples of radical innovations are invention of microwave, airplane etc.

In Harvard business review, the types of innovation are defined in terms of how well the problem is defined and how well the domain is identified and defined. Innovation types are (according to HBR): sustaining innovation, breakthrough innovation, disruptive innovation, and basic research. **Sustaining innovation** involves improving the existing capabilities. Organization tries to get better

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at what it has. Entrepreneur has a clear idea of what problems will come and which skill domains are needed to solve them. On the other hand, **breakthrough innovation** involves getting into a well-defined problem for which it is hard to get the solution. In this case, entrepreneur has to explore the skill domains which will be able to solve the problem. Researchers have discovered that with the changes in competition and shifts in technology and marketplace, companies improve and become better and better at the things (product/services) which customers want less. At this point, innovation in products or services won't help rather the companies have to innovate their business models. This type of innovation is called **disruptive innovation**. **Basic research** involves doing research on something you don't know. Every invention must have had a background research when the inventor did not even know what he is exploring.

Other most common types of innovation are: product innovation, business model innovation, process/technology innovation, social innovation, market-pull innovation and technology-push innovation. **Product innovation** involves innovation in products and intangible services such as services that meet customer needs. The business model is the way a company functions and earns money. **The business model innovation** encompasses innovations in strategy, marketing, supply chains, value creation, pricing or cost structures. The major examples of business model innovation are Food panda Uber, Careem, and Foodpanda. **Process and technology innovation** involves creating the process efficient or cost effective in order to improve quality or production. **Social innovations** are innovations where the benefit lies with society and the purpose is not primarily profit. Examples include innovation in education, poverty reduction, equal opportunities, health, pollution reduction, plastic use reduction, and green trends. **Market pull innovation** originates from the market and initiates by a specific customer request. Lastly, **technology-push innovation** is the result of new technologies for which suitable application possibilities are sought and implemented.

What is creativity?

As we know, creativity is what distinguishes humans from other creatures. Business managers and entrepreneurs are more interested in creativity as they think it as a linkage to innovation which ultimately leads to new product, process, firm growth, or stronger competitive advantage (Ko & Butler, 2007). Different academics define creativity as a process which results in novel or new products or ideas which are appropriate and valuable to the society as a whole. The traditional approach to creativity was person-centered which assumed that creativity is innate in individuals. Contrary to the traditional approach, the contemporary approach to creativity believes that all humans having normal capabilities can produce at least moderate creative work, for some time, and in some domain. Research showed that creativity is not only innate quality of an individual, but it is also the function of environmental factors and individual characteristics which can be developed (Amabile, 1996; Sternburg, 2006). Creativity is about seeing the things same as everyone else is seeing but making the connections between them which no one else has made. It's an ability to produce or use original, unusual and unique ideas. Business dictionary defines creativity as "mental

characteristics that allows a person to think outside of the box, which results in innovative or different approaches to particular task. There are the five stages of an Individual's creative process. Each stage of creative process flows logically into the next stage. The stages of creative process are: preparation, incubation, insight, evaluation, and elaboration.

As an individual starts the creative journey, the first phase involves **preparation** work and generation of idea. It's the stage when individual conducts the research work and gathers information and materials that could ignite a unique idea. In this first stage of the creative process, the brain is using its memory to get knowledge and information on past experiences to generate the novel idea. When an individual is done with actively thinking about the creative idea giving it upfront time, it is the stage when he/she must let it go for a while. This stage is called **incubation**. It is the part of creative thinking to get yourself away from your idea before sitting down again to add more details to it. At this stage an entrepreneur might start working on other project or take a break from creative thinking altogether in a sense that he/she is not actively or consciously working on his/her creative idea. Throughout this stage, the idea is incubating in the back of the mind. After incubation, entrepreneur enters into the **insight** stage which is sometimes called illumination stage, a stage when "aha" moment happens, and individual gets eureka experience (feeling of joy and satisfaction). It is when he is able to form new connections and gather all the material he has collected to present the solution to the problem. It is the stage when the real unique idea is emerged. At this stage, entrepreneur is sure about the target market, customers, the product, the sources of finance and resources. Once the idea is emerged, it enters in **evaluation** stage. It is the stage to validate the idea and weighing it against the alternatives. An individual can go back and see if the current version of his idea is what he initially imagined. Another important thing at this stage is testing the viability of the idea. The entrepreneur will have to scrutinize the idea (whether it is viable or not) and conduct market assessment. An important thing at this stage is to check the financial viability of the idea. **Elaboration** is the final stage of creative process, and is sometimes called verification stage. It is the stage when hard work is needed to transform your idea into a physical product or service. At this stage an entrepreneur finalizes the design, transform the idea into a physical product or service and share it with the world. Entrepreneur at this stage will launch the marketing campaigns, keep the financial records, take the customers' feedback and will try to improve based on the suggestions of the customers.

Barriers to Creativity and Innovation

Why an entrepreneur cannot innovate or bring creative product ideas into the market. There are certain barriers to innovation and creativity.

Personal barriers involves personality and behavior barriers, social pressure, lack of passion, lack of direction, fear of failure and making prior assumptions. *Personality and behavior* can be a major barrier to creativity and innovation. Some people are not the risk takers and want to enter in a market

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with a casual or simple (already existed) product. They might not have the capability to think out of the box. Thinking creatively or wanting to innovate is a personality trait too so the people who do not have that trait of doing something different, do not go for creativity. *Social pressure* is another barrier to a creative idea. An entrepreneur comes under the social pressure when he/she tells his/her idea to someone (family or friends) and gets a negative response (when they call your idea weird and strange). This is normal for friends, family and even customers to have innovation barriers in their minds which form a wall against creative things. Our brain spontaneously prefers the known things as compared to the unknown things. This is the reason why society first rejects your idea. Under this social pressure, even if the entrepreneur is creative, he/she will curtail his/her creativity and try to build a simple product which will have market acceptance. *Lack of passion* is a personal barrier which comes from demotivation. May be in the beginning, entrepreneur is excited about the new idea but after getting discouraged by the family and friends, he becomes less passionate about his creative idea. *Lack of direction* can also hinder an entrepreneur's creativity and innovation. If an entrepreneur's personality is inconsistent and he doesn't have a proper direction, it will lead him to astray. Only a focused person can think creatively. *Fear of failure* is the foremost barrier to innovation and creativity in entrepreneurship. It happens when an entrepreneur is already under the social pressure and people around him have already rejected his idea. It makes entrepreneur feel demotivated thinking that his product/service model will not work or be rejected by the customers too. This fear also disturbs the entrepreneur at market testing stage which can lead him/her to take wrong decisions. This fear of failure leads the entrepreneur to *make prior assumptions* about the product acceptability in the market. He assumes that he will not get the investment, resources, customers, or the team to work on the idea.

Cultural barriers involve lack of training, misguiding instructions, lack of motivation, lack of basic ability, and access to resources. Culture is defined as "the shared values, beliefs, and practices of the people". Some organizations do not have a creative culture. May be the business of the firm doesn't need creativity and innovation. Leg pulling and politics in organizations can also hinder creativity. In other firms (for example, advertising agencies or IT firms) creativity and innovation is the essential part of firm's culture. Such organizations (like Google) give upfront time to their employees to think creatively encouraging creative ideas and promoting innovation. An organization which needs creative ideas should train its employees to think creatively. *Lack of training* can be a barrier to creativity as the individual would not know what to think and what actually the organization needs him/her to think. Instructions on which type of idea is needed would be communicated clearly as *misguided instructions* can result in a totally different idea and wastage of time. Another barrier to creativity is *lack of motivation*. Even if the entrepreneur himself is creative but if he is unable to motivate others in the team, it will curb the creativity. If an entrepreneur doesn't have the *basic ability to think creatively* he/she won't be able to produce any unique ideas. Moreover, in order to execute a creative idea, entrepreneur needs resources. If he/she doesn't have *access to resources*, the creative idea will remain in the mind of the entrepreneur but will never be executed and offered in the market.

Feasibility barriers involve blur ideas, financial feasibility, and business model feasibility. Idea should be clear as the *blur idea* will not result in creative product or service. It should be practical and can be converted into a physical product or service. *Financial viability* is an important factor. The road from idea generation to a physical product/successful innovation is really expensive. An entrepreneur faces the situations like "we don't have the right staff", "too expensive", "this is practically impossible" etc. Most of the creative ideas die because the entrepreneur fails to meet the finances of the journey from idea generation to innovative product development.

The knowledge and experience barrier involves lack of prior knowledge, lack of prior experience, and lack of access to information. Prior knowledge of the domain and experience in the domain can enhance the creativity. According to Amabile (1996), one of the components of creativity is domain related experience. *Lack of prior knowledge and experience* of the domain can hinder the creativity. The more you have knowledge of the field, the more you think creatively. An entrepreneur should have updated information on new trends and business models. Inability to *access the right information* can curb the innovativeness and creativity.

The regulatory and procedural barrier involves laws and policies, registration process, access to raw materials and government policies. An entrepreneur has to execute his/her idea according to the rules, regulations, *laws and policies* of the country. He/she has to register the business as per the *registration process* of the relative country. Another major barrier can be the *access to raw material*. May be the idea is so unique that it needs raw materials which are hard to access or need governmental approval to access them. An entrepreneur also needs to get the patents for the creative idea and see the government policies regarding that. The idea must be acceptable in terms of *government laws and policies*. The product should not be something which is not allowed in a country where you want to offer it.

What makes entrepreneurs entrepreneurial? Another perspective on entrepreneurship

Effectuation: The problem

In entrepreneurship, the general question which comes in mind while thinking of starting a business startup is "How should we start a business?" Entrepreneurship experts state that the more important question is "How does a business start?" instead of "How should a business be started?" Another most discussed question in entrepreneurship is "**What makes entrepreneurs entrepreneurial?**" This debate led the researchers to give the concepts of causation and effectuation approach to entrepreneurship. Sara Sarasvathy is a researcher and expert in the field of entrepreneurship who gave the theory of effectuation in 2001. Before discussing theory of effectuation, it is important to understand the difference between causation approach to entrepreneurship and effectuation approach to entrepreneurship. A **causation approach** infers that "entrepreneurs focus on a predefined goal and then aim to find the means to reach this goal". Causal rationality arises when a pre-defined goals is set by the entrepreneur with a given set of means or resources. Entrepreneur

seeks to find the fastest, ideal, efficient and cheap means/alternatives to achieve that goal. The examples of causal reasoning are: choosing a high return market or choosing to serve the niche market. On the other hand, **an effectuation approach** infers that "entrepreneurs focus on the means at hand, which they aim to materialize into one or more goals that were not necessarily predefined." It does not start with a specific goal. Instead, effectuation starts with the given means that entrepreneur has at the moment. It allows goals to develop over time with the aspirations and imagination of the entrepreneur.

Difference between causation and effectuation process

Categories of differentiation	Causation process	Effectuation process
Practice	Start with goals	Start with their means: Who I am, what I know, whom I know
Logic	Predictive	Control
Money	Expected return	Affordable loss
Strategy	Competition	Partnerships
Perception	Exploitation of preexisting knowledge	Leveraging of contingencies

Source: Based on Sarasvathy (2001a, 2001b, 2003).

Effectuation: The Process

As we know, effectuation does not start with a specific goal. Instead, effectuation starts with the given means that entrepreneur has at the moment. All entrepreneurs begin with three categories of means:

(1) **Who they are** – their traits, tastes and abilities; (some people have certain special skills or qualities, natural skills and abilities which are actually the resources. Resources are not only the financial resources rather a special skill or ability can also be a resource).

(2) **What they know** – their education, training, expertise, and experience; (for example, if you have education in data sciences, you can start your own IT firm based on your education and expertise as this education and expertise are your resources/means)

(3) **Whom they know** – their social and professional networks (networking with resourceful and useful people can also be a resource/mean for an entrepreneur)

These three things are the core of an entrepreneur as entrepreneurs use these things (traits, education, skills, expertise and experience) to generate business ideas and start their businesses.

Effectuation: The Principles

There are five principles of effectuation given by Dr. Sarasvathy which are:

1. Bird in hand principle (create solutions to the problem with the means/resources available at the moment)
2. Lemonade principle (surprises and mistakes are unavoidable. An entrepreneur should use the mistakes and surprises to find new opportunities.)
3. Crazy quilt principle (focus on co-creation and partnership instead of competition. It can bring new directions and funds for your project.)
4. Affordable loss principle (you should see how to cover the negative side. The best approach is to invest only the amount which you can afford to lose.)
5. Pilot-in-the-plane principle (you are yourself responsible for everything. Your future can be influenced by your own actions.)

Bird in Hand Principle

When successful entrepreneurs start a new business, they first start with the available resources or means. The bird in hand principle infers that the business should be started with the means/resources currently available. These resources are categorized as follows:

- Who I am- what skills, abilities and tastes an entrepreneurs has
- What I know- his/her education, experience and expertise
- Who I know- his/her professional and social network

Using the mixture of these means, an entrepreneur can start imagining the new possibilities. Usually, he/she starts a business at a small scale with closest available resources or means and directly implements it without any detailed planning. With each decision, possible results are re-aligned. Ultimately, certain emerging things result in clearly achievable goals. The end goals are the result of aspirations, abilities, skills, and imagination of the entrepreneur and the people he/she has interacted with.

Affordable Loss Principle

In business world, the general practice is to choose the market with highest return. It is a natural impulse which comes from the years long training which we get in business schools and business world. Business world tells us to maximize the returns by choosing optimal strategy and high return market segment. As far as effectual entrepreneurs are concerned, they go for affordable loss instead of maximizing profits. They make decisions on what they can afford to lose not what they can expect to make. This is what affordable principle infers. Instead of calculating what money is needed upfront and investing his/her time, energy and effort in raising that money, an expert entrepreneur tries to assess the negative side or downside and examines what he/she is willing to lose. The entrepreneur then starts the process of building the venture and brings other partners or stakeholders on board in order to estimate what they can afford to lose. An estimation of affordable

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loss does not depend on what type of venture an entrepreneur is going to start rather it depends on the entrepreneur himself. It varies from one person to the other and even across the life stage of the entrepreneur and circumstances he/she is in. By estimating affordable loss and making decision which business to start (based on estimation of affordable loss), entrepreneurs stop relying on prediction. As an alternative, they cultivate the opportunities that have a low cost of failure.

Crazy Quilt Principle

While causation relies on detailed competitive analysis, effectuation focuses on co-creation and partnership. The crazy quilt principle focuses on building partnerships instead of beating competition. Since effectual entrepreneurs start their business without a predetermined target market, the detailed competitors' analysis becomes less useful for them as they do not have an idea of who will be their competitors. Instead, they take their product to nearest possible potential customers. Some of the individuals or firms they interact with make commitments with the entrepreneurs by investing their means/resources in terms of time, efforts and money. This partnership principle or crazy quilt principle merges best with affordable loss principle to bring entrepreneur's product into the market with little expenditure. In the initial stages of venture development, these partnerships with suppliers, customers and other stakeholders help reduce the uncertainty and fear of failure. Lastly, since in the beginning entrepreneur doesn't not know which market to serve, these strategic partnerships and network, to a great extent, determine which market/markets the entrepreneur will eventually target.

Lemonade Principle

There is a famous proverb "If life deals you lemons, make lemonade!" The lemonade principle (leverage contingences) of effectuation infers that an entrepreneur should embrace surprises that arise from uncertain circumstances by being flexible instead of being tied to the set goals. This principle is at the heart of entrepreneurial expertise that is the ability to turn the surprises or unexpected things into profitable opportunities. Effectual entrepreneurs learn to work with uncertainties and take advantage from them. In most businesses, the contingency plans are prepared where surprises are considered worst case scenarios. On the contrary, because effectual entrepreneurs do not bind their idea to any preconceived or specific market, so everything and anything is a surprise for them which can be turned into a valuable opportunity.

Pilot-in-the-Plane Principle

The desire of self-control is as old as human itself is. Research shows that humans have been struggling to control everything in their lives ever since the history began. In today's world, the mechanism, processes and instruments for control-striving are different than they were in the past but the issue remains the same. Psychological research studies show that human behavior of control-striving is directly related to his/her functioning/performance. For example, if a person has self-control or control over things around him, he or she will be less stressed. On the other hand, loss of control is linked with helplessness and depression. Effectual entrepreneurs consider the desire of control as healthy and normal. Entrepreneurs usually chose entrepreneurship to become their own boss and control his/her business. Control makes them able to work independently, and on their own schedule. Effectual entrepreneurs take right actions, at the right time to control the things. They focus on the things in their own control and know that their decisions will result in the outcome they want. Though every entrepreneur has the desire to control but the intensity of it varies among people and also depends on the life stage of the entrepreneur. This principle suggests that entrepreneur is similar to a pilot who is responsible for the plane and passengers just like the entrepreneur is responsible for his venture.

----------The End----------

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